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Report

Project2

David Albino[1], **Diogo Centeno**[2], **Mariana Galego**[3], **Nuno Wang**[4] **Shanjun Yang**[5]
[1] 105993, MMAC [2] 102822, LMAC [3] 106967, MMAC [4] 106343, LMAC [5] 117310, MEEC

1 Introduction

Non-Revenue Water (NRW) and Real Losses (RL) are key indicators of the efficiency and sustainability of water supply systems, playing a central role in regulatory oversight and performance benchmarking. However, traditional percentage-based measures and fixed performance thresholds often fail to account for structural heterogeneity across utilities, limiting comparability and interpretability.

This study analyses alternative intensity-based indicators for NRW and Real Losses using regulatory data from the Portuguese water sector. The proposed indicators explicitly account for system scale, network characteristics, and service structure, enabling more meaningful cross-utility comparisons. A set of structural, operational, and governance-related variables is examined to identify the main drivers of observed performance differences.

A central focus of the analysis is the classification of utilities into performance categories. Several empirical, quantile-based classification schemes are evaluated and compared with ERSAR's official performance classes. The results highlight systematic limitations of purely relative thresholds, particularly in contexts of widespread underperformance. To address these limitations, a hybrid classification framework is proposed, combining regulatory anchoring with data-driven discrimination. This approach aims to improve interpretability, stability, and policy relevance while preserving statistical robustness.

2 Indicators for NRW and its components

The assessment of non revenue water and its components has been widely discussed in the literature and by international organizations such as the International Water Association (IWA). Several indicators have been proposed to overcome the limitations of simple volumetric or percentage based measures and to account for differences in network size and system characteristics.

The Portuguese water regulator, ERSAR, assesses non revenue water and its components using indicators that are designed to ensure regulatory comparability across utilities operating under diverse conditions. ERSAR quantifies NRW as a percentage of system input volume, providing a simple and intuitive measure of the share of water that does not generate revenue. However, it has increasingly been recognized that it's not an appropriate indicator to benchmark NRW levels between utilities or even to monitor changes over time, because percentage based indicators do not account for network structure, system size or service density. As a result, utilities with very different physical and operational characteristics may exhibit similar percentage values, limiting comparability.

This problem can be mitigated by measuring NRW not as a share, but in terms of absolute losses per connection per day or per mains length per day. The International Benchmarking Network for Water Sanitation recommends to use different indicators, such as percentage, losses per connection or losses per km of network together. While normalizing per connection is more suitable for urban systems with high density and may overstate in rural areas, normalizing per mains length is more suitable for rural systems with low connection density. This reasoning aligns with the ERSAR indicator for real losses that adopts an intensity based approach computing the two alternative indicators, real losses per mains length ($m^3/km.day$) and real losses per service connection ($L/connection.day$). The selection of the indicator used for performance assessment depends on the type of system, bulk or retail, and on connection density, with the length based indicator applied to bulk and low density systems and the connection based indicator applied to high density retail systems.

In addition to real losses, apparent losses and authorized unbilled consumption (AUC) can be assessed using similar approaches. Apparent losses include metering inaccuracies or data handling errors and AUC corresponds to water supplied but not billed and either can be expressed as a percentage of system input volume or normalized by structural characteristics, such as the number of service connections or mains length. The IWA and other performance frameworks including ERSAR, recommend using both volumetric and normalized indicators to capture relative significance of these components.

$$\text{UA_vol} = \text{V_Consumo_não_faturado_medido} + \text{V_Consumo_não_faturado_não_medido}$$

$$\text{UA_new} = \begin{cases} \dfrac{\text{UA_vol} \times 1000}{\text{V_Ramais_de_ligação} \times 365}, & \text{if Tipo_Sistema} = \text{"Retail" \& I_Densidade_de_ramais} \geq 20 \\ \dfrac{\text{UA_vol}}{\text{V_Comprimento_total_de_condutas} \times 365}, & \text{if Tipo_Sistema} = \text{"Bulk"} \\ & \text{or Tipo_Sistema} = \text{"Retail" \& I_Densidade_de_ramais ¡ 20} \end{cases}$$

$$\text{AL_vol} = \text{V_Uso_não_autorizado} + \text{V_Perdas_de_água_por_erros_de_medição}$$

$$\text{AL_new} = \begin{cases} \dfrac{\text{AL_vol} \times 1000}{\text{V_Ramais_de_ligação} \times 365}, & \text{if Tipo_Sistema} = \text{"Retail" \& I_Densidade_de_ramais} \geq 20 \\ \dfrac{\text{AL_vol}}{\text{V_Comprimento_total_de_condutas} \times 365}, & \text{if Tipo_Sistema} = \text{"Bulk"} \\ & \text{or Tipo_Sistema} = \text{"Retail" \& I_Densidade_de_ramais ¡ 20} \end{cases}$$

Although ERSAR computes two intensity based indicators for real losses, only one is used for performance classification, depending on density. This conditional selection restricts comparability across utilities assessed using different normalizations and may obscure situations where utilities exhibit similar overall loss intensity but rank differently due to the chosen unit. To attempt on finding a more balanced measure on real losses, one might use the Combined Real Loss Index, CRLI, that represents the geometric mean of real losses per connection per day ($L/connection.day$) and real losses per mains length ($L/km.day$). This indicator aims to integrate both infrastructure exposure and service intensity and has been applied in benchmarking platforms such as AquaBench.

$$CRLI = \sqrt{\text{Real losses per connection per day} \times \text{Real losses per mains length}}$$

Another concern of ERSAR's indicator for real losses is that it does not explicitly relate observed real losses to unavoidable or minimum achievable levels. To address this limitation, there is another indicator, Infrastructure Leakage Index, ILI, that IWA uses as a performance indicator for leakage which adjusts the measured loss, taking into account the service pressure and the length of the network. The ILI is computed as follows:

$$\text{ILI} = \frac{\text{CARL}}{\text{UARL}}$$

where CARL represents the Current Annual Real Losses and UARL represents the Unavoidable Annual Real Losses. UARL corresponds to the lowest technically achievable annual volume of real losses for well maintained and well managed systems. Each system has a different UARL base level, which varies widely depending on density of connections, length of connections and average pressure. System-specific values of UARL can be assessed using a formula developed by the IWA Water Losses Task Force (1999). Data required for this assessment are the number of service connections (N_c), the length of mains (L_m), the length of private pipes (L_p) and the average operating pressure (P). The general equation for UARL is

$$\text{UARL (L/day, when system is pressurised)} = (18 \times L_m + 0.8 \times N_c + 25 \times L_p) \times P$$

This equation, based on component analysis of real losses for well managed systems with good infrastructure, has proved to be robust in diverse international situations and is the most reliable predictor yet of 'how low could you go' with real losses for systems with more than 5000 service connections, connection density more than 20km and average pressure more than 25 metres.

Since 1999, lower and upper limits on system size and average pressure for use of the UARL equation have been cautiously changed and a non dimensional multiplier called System Correction Factor can be applied to the UARL equation to customise UARL volume for smaller systems or for average pressures outside the allowed range.

3 Proposed Indicators and Explanatory Variables

3.1 Proposed indicators for NRW and RL

Based on the review of international best practices and regulatory approaches presented in the previous section, this study proposes a set of performance indicators for non revenue water and real losses that aim to balance interpretability, comparability and data availability within the scope of ERSAR's data.

For real losses, different indicators are adopted according to system type. In retail systems, real losses will be measured using CRLI. This indicator could capture both service intensity and infrastructure exposure, allowing for a more balanced comparison across retail systems. In bulk systems, where service connections are absent, real losses will be assessed using real losses per mains length ($m^3/km \cdot day$), which already is the indicator used by ERSAR.

For non revenue water two intensity based indicators will be considered: NRW per mains length ($m^3/km \cdot day$) and NRW per service connection ($L/connection \cdot day$). Consistent with ERSAR's methodological approach, the selection of the indicator used for analysis will depend on system type and connection density. High density ($\geq 20km$) retail systems will be evaluated using the per connection indicator, while low density retail systems and bulk systems will be assessed using the length based indicator. This conditional selection ensures that the NRW indicator reflects the dominant structural characteristics for each system.

3.2 Analysis of correlation between proposed indicators and candidate explanatory variables

For analytical purposes, these indicators were combined into a single variable (`NRW_new`), which assigns to each utility the indicator appropriate to its system type and connection density. The corresponding unit is retained in a separate variable, ensuring that comparisons, classifications and thresholds are applied consistently within each regime. Similarly, the same procedure was conducted for `RL_new`.

The identification of explanatory variables for the proposed NRW and real loss indicators was guided by both theoretical considerations and data availability within ERSAR's regulatory dataset. Variables were selected to reflect four main dimensions commonly highlighted in the literature: system scale, operational performance, infrastructure condition and governance characteristics.

An initial exploratory analysis was conducted using pairwise correlations between the proposed indicators and candidate explanatory variables. This exploratory step was used to identify potential associations and to guide the subsequent model specification, rather than to determine variable inclusion mechanically.

	y	x	cor	n
1	NRW_new	I_Ocorrência_de_falhas_no_abastecimento	0.61604256	206
2	NRW_new	V_Indice_de_conhecimento_infraestrutural	-0.21602044	206
3	NRW_new	V_Perdas_reais	0.18562558	206
4	NRW_new	V_Indice_de_gestão_patrimonial_de_infraestruturas	-0.17661809	206
5	NRW_new	I_Adesão_ao_serviço	0.15369664	206
6	NRW_new	V_Pessoal_em_outsourcing_afeto_ao_serviço_de_abastecimento_de_água	-0.13186360	206
7	NRW_new	V_Comprimento_total_de_condutas	-0.12843168	206
8	NRW_new	V_Consumo_não_faturado_não_medido	0.12791348	206
9	NRW_new	V_Comprimento_médio_de_condutas_com_mais_de_10_anos	-0.12609402	206
10	NRW_new	V_Falhas_no_abastecimento	-0.11101505	206
11	NRW_new	V_Água_faturada	-0.11054510	206
12	NRW_new	V_Gastos_totais	-0.10237268	206
13	NRW_new	V_Água_entrada_no_sistema	-0.08821338	206
14	NRW_new	I_Ramais_afetados_por_falhas_no_abastecimento	0.07457010	196
15	NRW_new	V_Ramais_afetados_por_falhas_no_abastecimento	0.07176529	196
16	NRW_new	I_Densidade_de_ramais	0.07021568	196
17	NRW_new	conn_density	0.06895377	196
18	NRW_new	V_Rendimentos_tarifários	-0.06843433	206
19	NRW_new	V_Ramais_de_ligação	-0.06764954	196
20	NRW_new	I_Reabilitação_de_condutas	0.06754415	206

(a) Correlation between `NRW_new` and candidate explanatory variables

	y	x	cor	n
1	RL_new	I_Ocorrência_de_falhas_no_abastecimento	0.53059414	206
2	RL_new	V_Indice_de_conhecimento_infraestrutural	-0.27482865	206
3	RL_new	V_Indice_de_gestão_patrimonial_de_infraestruturas	-0.23838556	206
4	RL_new	V_Comprimento_total_de_condutas	-0.18823614	206
5	RL_new	V_Comprimento_médio_de_condutas_com_mais_de_10_anos	-0.18381702	206
6	RL_new	V_Perdas_reais	0.17036215	206
7	RL_new	I_Densidade_de_ramais	0.14826306	196
8	RL_new	conn_density	0.14758636	196
9	RL_new	V_Pessoal_em_outsourcing_afeto_ao_serviço_de_abastecimento_de_água	-0.14477336	206
10	RL_new	V_Água_faturada	-0.14471077	206
11	RL_new	V_Gastos_totais	-0.13875864	206
12	RL_new	V_Falhas_no_abastecimento	-0.12810560	206
13	RL_new	V_Ramais_de_ligação	-0.12616861	196
14	RL_new	V_Água_entrada_no_sistema	-0.12443477	206
15	RL_new	I_Adesão_ao_serviço	0.12256269	206
16	RL_new	V_Rendimentos_tarifários	-0.10802478	206
17	RL_new	I_Ramais_afetados_por_falhas_no_abastecimento	0.05772933	196
18	RL_new	V_Ramais_afetados_por_falhas_no_abastecimento	0.05733207	196
19	RL_new	V_Consumo_não_faturado_medido	-0.05582174	206
20	RL_new	V_Condutas_reabilitadas_nos_últimos_cinco_anos	-0.03476462	206

(b) Correlation between `RL_new` and candidate explanatory variables

Figure 1: Correlations between proposed indicators and candidate explanatory variables

The correlation analysis indicates that Real Losses are most strongly associated with indicators reflecting operational reliability and infrastructure condition. The highest positive correlation is observed with the occurrence of failures in water supply, suggesting that systems experiencing more frequent failures tend to exhibit higher real losses. Connection density also shows a positive association, indicating that denser networks may face greater challenges in controlling physical losses. In contrast, network scale variables such as total network length and the proportion of older pipes display negative correlations, consistent with the presence of scale effects whereby larger systems may achieve lower real losses per unit. Financial and demand-related variables (e.g. billed water, total costs, and system input) show weaker correlations, suggesting that physical and operational characteristics play a more relevant role in explaining real losses than purely economic factors.

For the proposed NRW indicator, the strongest correlation is observed with the occurrence of failures in water supply, highlighting the close link between operational disruptions and overall water losses. NRW also shows moderate associations with indicators related to infrastructure management and customer impact, such as the number of affected connections and consumption not billed or not metered. Variables related to network size and system input exhibit weaker correlations, suggesting that NRW is not driven solely by scale but by a combination of operational performance and infrastructure condition. Overall, the results indicate that NRW captures a broader set of loss mechanisms than Real Losses, encompassing both physical losses and commercial or administrative inefficiencies, which is consistent with its conceptual definition.

3.3 Identification of explanatory variables for NRW and Real Losses

The purpose of this section is to identify explanatory variables associated with the proposed indicators of NRW and Real Losses. The emphasis is placed on explanation rather than prediction: the objective is to understand which structural, operational, and contextual factors are systematically related to water losses and to assess the robustness and interpretability of these relationships.

Several linear regression models were estimated for both indicators. For clarity and conciseness, all tested models are briefly described, while detailed interpretation is provided only for a baseline specification and for the final selected models, which display the most satisfactory diagnostic behaviour.

3.3.1 Explanatory variables for NRW

The analysis focused on operational failures, network scale, system throughput, management model, system type, and area typology, which are all theoretically relevant drivers of NRW through physical losses, apparent losses, and operational inefficiencies. Even though real losses volume and non billed authorized consumption showed moderate correlations with the indicators, their correlation reflects a denitional relationship rather than an independent explanatory effects. For this reason, the only volume metric used in the models is the system input volume, because it is widely recognised as a key explanatory variable reflecting system scale and demand intensity.

Additionally, variables related to service connections were excluded from the final models, because these variables are not defined for bulk water systems and therefore would bias the analysis towards retail systems. Overall, variable selection was guided by correlation analysis, conceptual consistency, multivariate significance and data availability across both bulk and retail utilities.

The model for NRW (`m1`) relates `NRW_new` to fundamental structural and contextual variables, namely infrastructure knowledge, failure occurrence, network length, system input volume, management model, system type, and area typology, while also allowing for an interaction between failure occurrence and network length. This specification exhibits a high explanatory power, with an R^2 of approximately 0.73, indicating that these variables capture important structural and operational drivers. In particular, failure occurrence and system input volume exhibit strong positive associations with NRW, while network length displays a negative main effect that is partially offset by the interaction term, suggesting diminishing marginal effects of failures in larger networks.

Diagnostic checks indicate that the model satisfies some core assumptions of the linear regression framework. No evidence of serial correlation in the residuals is detected and variance inflation factors remain within acceptable ranges. However, residual diagnostics reveal the presence of heteroskedasticity and some deviation from normality, as shown in Figure 2, with greater dispersion observed for higher values of NRW. These features are consistent with the scale and heterogeneity of water loss indicators and do not suggest fundamental misspecification of the mean structure of the model.

```
Call:
lm(formula = NRW_new ~ V_Indice_de_conhecimento_infraestrutural +
    I_Ocorrência_de_falhas_no_abastecimento + log_len + log_in +
    I_Modelo_de_gestão + V_Tipologia_da_área_de_intervenção +
    Tipo_Sistema + I_Ocorrência_de_falhas_no_abastecimento *
    log_len, data = model_df)

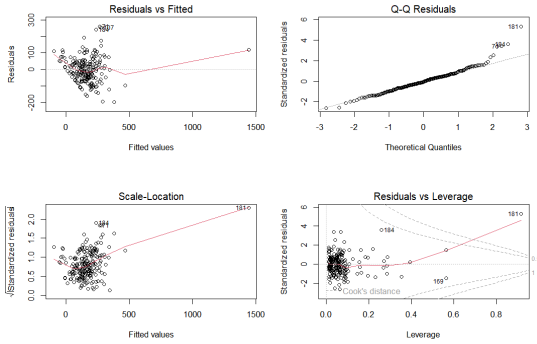
Residuals:
     Min       1Q   Median       3Q      Max
-198.120  -49.012   -3.607   42.230  258.927

Coefficients:

                                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                                  -1684.8370   174.6603  -9.646  < 2e-16 ***
V_Indice_de_conhecimento_infraestrutural                        -0.2747     0.1656  -1.659 0.098759 .
I_Ocorrência_de_falhas_no_abastecimento                         62.5856     5.6172  11.142  < 2e-16 ***
log_len                                                        -72.6474    10.9602  -6.628 3.38e-10 ***
log_in                                                         124.7109    12.2365  10.192  < 2e-16 ***
I_Modelo_de_gestãoConcessão (concessão municipal)              131.1233    47.3884   2.767 0.006214 **
I_Modelo_de_gestãoDelegação (empresa estatal)                  -21.3662    66.5901  -0.321 0.748666
I_Modelo_de_gestãoDelegação (empresa municipal ou intermunicipal) 130.3270  45.8286   2.844 0.004943 **
I_Modelo_de_gestãoDelegação (parceria Estado/ municípios)       69.9032    55.9244   1.250 0.212844
I_Modelo_de_gestãoGestão direta (serviço municipal)            224.1258    47.0528   4.763 3.76e-06 ***
I_Modelo_de_gestãoGestão direta (serviço municipalizado ou intermunicipalizado) 184.9136 48.4256 3.819 0.000181 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente rural  67.3988    17.6885   3.810 0.000187 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente urbana -16.3511    21.8739  -0.748 0.455673
Tipo_SistemaBaixa                                              258.9249    43.7312   5.921 1.46e-08 ***
I_Ocorrência_de_falhas_no_abastecimento:log_len                  -9.6093     1.1562  -8.311 1.74e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 77.58 on 191 degrees of freedom
Multiple R-squared:  0.7327,    Adjusted R-squared:  0.7131
F-statistic: 37.39 on 14 and 191 DF,  p-value: < 2.2e-16
```

(a) `m1` summary



(b) `m1` plots

Figure 2: Model `m1`

To address this issue, alternative model specifications using transformations of the dependent variable, such as logarithmic and square root transformations were explored. Results are shown in section 8.1. These transformations improved the behaviour of the residuals and partially mitigated heteroskedasticity, but did not materially alter the set of explanatory variables or the direction of estimated effects.

Instead, to ensure reliable inference, robust standard errors were computed while keeping the indicator in its original units. This approach preserves the interpretability of the coefficients in terms of the proposed indicators while providing reliable inference under heteroskedasticity. The estimates and the resulting p-values are therefore used to identify explanatory variables in a manner consistent with the scope and objectives of the study and are displayed in figure 3.

```
t test of coefficients:

                                                                Estimate  Std. Error t value  Pr(>|t|)
(Intercept)                                                  -1684.83696  188.58067 -8.9343 3.454e-16 ***
V_Indice_de_conhecimento_infraestrutural                        -0.27466    0.15576 -1.7634  0.079434 .
I_Ocorrência_de_falhas_no_abastecimento                         62.58562    9.50981  6.5812 4.379e-10 ***
log_len                                                        -72.64742   10.70897 -6.7838 1.428e-10 ***
log_in                                                         124.71092   13.45024  9.2720 < 2.2e-16 ***
I_Modelo_de_gestãoConcessão (concessão municipal)              131.12328   42.98621  3.0504  0.002611 **
I_Modelo_de_gestãoDelegação (empresa estatal)                  -21.36618   68.47320 -0.3120  0.755353
I_Modelo_de_gestãoDelegação (empresa municipal ou intermunicipal) 130.32696 41.35000 3.1518  0.001884 **
I_Modelo_de_gestãoDelegação (parceria Estado/ municípios)       69.90317   37.12121  1.8831  0.061206 .
I_Modelo_de_gestãoGestão direta (serviço municipal)            224.12584   44.54763  5.0312 1.121e-06 ***
I_Modelo_de_gestãoGestão direta (serviço municipalizado ou intermunicipalizado) 184.91356 45.98558 4.0211 8.334e-05 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente rural  67.39879   20.78465  3.2427  0.001397 **
V_Tipologia_da_área_de_intervençãoArea predominantemente urbana -16.35107  27.53045 -0.5939  0.553264
Tipo_SistemaBaixa                                              258.92488   38.00979  6.8121 1.219e-10 ***
I_Ocorrência_de_falhas_no_abastecimento:log_len                  -9.60931    2.00359 -4.7961 3.250e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 3: `m1` summary with robust standard errors

Overall, the results of the model `m1` identify a clear set of explanantory variables associated with the proposed non revenue water indicator. Operational reliability, proxied by `I_Ocorrência_de_falhas_no_abastecimento`, is a strong positive driver of NRW. System input volume (`log_in`) and network length (`log_len`), introduced in logarithmic forms to capture scale effects, exhibit main opposite effect. Whereas, larger volumes flowing through the system increase exposure to both real and apparent losses, larger networks tend to display lower NRW intensity levels. The use of logarithmic transformations for scale related variables allows non linear relationships to be captured while preserving interpretability in terms of relative changes rather than absolute measures. Moreover, the statistically significant interaction between failure occurrence and network length indicates that the marginal impact of failures on NRW decreases as network size increases. Contextual and organisational characteristics also contribute to explaining NRW variability. Management model and system type are systematically associated with NRW, highlighting the influence of institutional arrangements and the distinction between bulk and retail systems. Area typology further indicates that predominantly rural systems tend to exhibit higher NRW levels. While not all category levels within management model and area typology are individually statistically significant relative to the reference categories, these variables are retained due to their theoretical relevance and their overall contribution to explaining systematic variation in NRW across utilities. Taken together, the analysis identifies failure occurrence, system

input volume, network scale, system type, management model and area typology as a set of explanatory variables for NRW within the scope of the available ERSAR data.

3.3.2 Explanatory variables for Real Losses

A similar modelling approach is adopted for the Real Losses indicator.

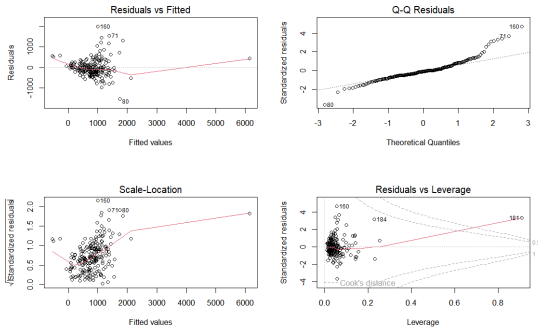
```
Call:
lm(formula = RL_new ~ V_Indice_de_conhecimento_infraestrutural +
    I_Ocorrência_de_falhas_no_abastecimento + log_len + log_in +
    log(V_Rendimentos_tarifários) + I_Adesão_ao_serviço +
    Tipo_Sistema + V_Tipologia_da_área_de_intervenção + I_Ocorrência_de_falhas_no_abastecimento *
    log_len, data = model_df)

Residuals:
     Min       1Q   Median       3Q      Max
-1548.10  -216.05   -35.35   172.87  1991.66

Coefficients:
                                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                                             -4800.2178   885.9866  -5.418 1.76e-07 ***
V_Indice_de_conhecimento_infraestrutural                   -2.9408     0.9014  -3.263   0.0013 **
I_Ocorrência_de_falhas_no_abastecimento                   202.7982    31.0287   6.536 5.40e-10 ***
log_len                                                  -480.4506    65.9730  -7.283 7.89e-12 ***
log_in                                                    784.3922    81.1221   9.669  < 2e-16 ***
log(V_Rendimentos_tarifários)                            -271.4838    60.6570  -4.476 1.29e-05 ***
I_Adesão_ao_serviço                                        -7.9839     4.4292  -1.803   0.0730 .
Tipo_SistemaBaixa                                        1905.8854   200.3017   9.515  < 2e-16 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente rural   231.1834    94.4570   2.447   0.0153 *
V_Tipologia_da_área_de_intervençãoArea predominantemente urbana  -47.8371   121.4046  -0.394   0.6940
I_Ocorrência_de_falhas_no_abastecimento:log_len           -28.9686     6.4103  -4.519 1.08e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 438 on 195 degrees of freedom
Multiple R-squared:  0.6334,    Adjusted R-squared:  0.6146
F-statistic: 33.69 on 10 and 195 DF,  p-value: < 2.2e-16
```

(a) m4 summary



(b) m4 plots

Figure 4: Model m4

The initial linear specification for real losses (m4) relates the indicator RL_new to key structural and operational variables, including infrastructure knowledge, failure occurrence, network scale, system throughput, tariff revenue, service adherence, management-related factors, system type, and area typology, while also allowing for an interaction between failure occurrence and network length. This model explains a substantial share of the variability in real losses ($R^2 \approx 0.63$), indicating that physical and operational characteristics are key drivers of leakage-related losses.

Operational reliability (I_Ocorrência_de_falhas_no_abastecimento) and system input volume (log_in) are positively associated with the indicator, reflecting that higher failure frequencies and higher volumes fo water increase exposure to losses. Network length (log_len) exhibits a negative main effect, partially offset by the significant interaction with failure occurrence, suggesting diminishing marginal impacts of failures in larger networks. Financial and service related variables plays a meaningful to moderate role. Higher tariff revenues (log(V_Rendimentos_tarifários)) are associated with lower indicator values, consistent with better resourced systems being able to mitigate losses. Service adherence (I_Adesão_ao_serviço) shows a weak negative effect overall. Contextual variables also influence outcomes. Predominantly rural areas tend to have higher indicator values, while urban areas do not differ significantly from the reference category. System type matters as well: retail systems exhibit substantially higher values than bulk systems, reflecting both operational and reporting differences.

Diagnostic analysis of the linear model indicates the presence of heteroskedasticity as well as departures from normality in the residual distribution. Variance inflation factors remain below conventional thresholds, indicating no problematic multicollinearity, even in the presence of interaction terms. The Durbin- Watson test suggests mild positive autocorrelation. However, given the cross-sectional nature of the data and the absence of an inherent temporal or heterogeneity rather than true serial dependence.

To address these limitations, a log-transformed model (m41 can be found in 8.2) is estimated with log(RL_new) as the dependent variable. This specification achieves a very high adjusted R^2 (approximately 0.87) and substantially better diagnostic behaviour, with more homoscedastic residuals and reduced influence of extreme observations. The results consistently indicate that operational failures, system throughput, network scale, financial and service related variables and system type are key explanatory variables for real losses. However, it also implied modelling a transformed version of the indicator, which shifts the interpretation of the coefficients away from the original performance metrics.

Therefore, following a similar approach as the one used for the NRW indicator, robust standard errors were used while keeping the indicator in its original units. The estimates and the resulting p-values are therefore used to identify explanatory variables in a manner consistent with the scope and objectives of the study and are displayed in figure 5.

```
t test of coefficients:

                                                                Estimate Std. Error t value  Pr(>|t|)
(Intercept)                                                  -4800.2178   956.7395 -5.0173 1.176e-06 ***
V_Indice_de_conhecimento_infraestrutural                        -2.9408     1.0254 -2.8679 0.0045871 **
I_Ocorrência_de_falhas_no_abastecimento                        202.7982    41.3533  4.9040 1.973e-06 ***
log_len                                                       -480.4506    90.3084 -5.3201 2.831e-07 ***
log_in                                                         784.3922    99.8009  7.8596 2.549e-13 ***
log(V_Rendimentos_tarifários)                                 -271.4838    72.6850 -3.7351 0.0002463 ***
I_Adesão_ao_serviço                                             -7.9839     4.6478 -1.7178 0.0874244 .
Tipo_SistemaBaixa                                             1905.8854   224.3602  8.4948 5.045e-15 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente rural  231.1834    93.0321  2.4850 0.0137975 *
V_Tipologia_da_área_de_intervençãoArea predominantemente urbana -47.8371   137.5582 -0.3478 0.7283960
I_Ocorrência_de_falhas_no_abastecimento:log_len                 -28.9686     9.0874 -3.1878 0.0016703 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 5: m4 summary with robust standard errors

An additional model incorporating a subsample estimation by system type was also explored (m4b can be found in 8.2). This was mainly done to assess the explanatory power of variables related to service connections, namely I_Densidade_de_ramais. While some variables are statistically significant in this specification, the overall gain in explanatory power is limited, and diagnostic issues re-emerge in several cases.

Overall, the results of the model identify a coherent set of explanatory variables associated with the proposed real losses indicator. Operational reliability, proxied by I_Ocorrência_de_falhas_no_abastecimento, is a strong positive driver of real losses, with its marginal impact decreasing as network length (log_len) increases, as indicated by the statistically significant interaction term. System input volume (log_in) also emerges as a positive determinant, while larger network length is associated with lower real losses, highlighting the scale effects captured through the logarithmic transformations. Financial and service-related variables, including tariff revenues (log(V_Rendimentos_tarifários)) and service adherence (I_Adesão_ao_serviço), contribute additional explanatory power, with higher revenues mitigating real losses. Contextual characteristics, such as system type and area typology, further shape performance: low-complexity systems exhibit substantially higher real losses, and predominantly rural areas show modestly elevated losses compared to other typologies. While not all category levels are individually statistically significant, they are retained for their theoretical relevance and their role in capturing systematic variation across utilities. Collectively, the analysis identifies infrastructure knowledge index, failure occurrence, network length, system water volume, tariff revenue, service adherence, type of system, management model and area typology as a set of explanatory variables for the proposed real losses indicator withing the scope of ERSAR's data.

A comparison between the explanatory variables identified for non revenue water and real losses reveals a partial overlap, reflecting the fact that a substantial share of NRW in the dataset is attributed to real losses. Variables capturing operational reliability, such as the occurrence of supply failures and structural characteristics, including network length and system input volume are consistently relevant for both indicators. System type, area typology and management model also contribute to explaining variation in both NRW and real losses. This overlap highlights that while NRW encompasses apparent losses, real losses and unbilled authorized consumption, the patterns driving real losses are a dominant component within the observed NRW.

4 Classification and Comparison

Given the diversity of system types, indicator units, and the strongly skewed empirical distributions observed for both Non-Revenue Water (NRW) and Real Losses (RL), a relative, data-driven classification approach was adopted. Rather than relying exclusively on predefined absolute benchmarks, utilities were classified according to their position within the empirical distribution of sector performance, computed separately for each indicator and unit, ensuring comparability across heterogeneous utilities, robustness to extreme values and adaptability to structural changes in sector performance over time.

To evaluate the sensitivity of the classification to different distributional assumptions, multiple quantile-based schemes were tested and systematically compared against the official ERSAR performance classes.

4.1 Quantile-based classification schemes

Four quantile-based schemes were considered:

- Terciles: thresholds at the 33rd and 67th percentiles
- Quartiles: thresholds at the 25th and 75th percentiles

- p20–p80: thresholds at the 20th and 80th percentiles
- p40–p80: thresholds at the 40th and 80th percentiles

Thresholds were computed separately for each unit of measurement to avoid mixing structurally different indicators. For each scheme and each indicator, utilities were classified as:

$$\text{Class} = \begin{cases} \text{Good}, & \text{if } x_i \leq c_1 \\ \text{Medium}, & \text{if } c_1 \leq x_i \leq c_2 \\ \text{Unsatisfactory}, & \text{if } x_i \geq c_2 \end{cases}$$

where x_i represents the value of the performance indicator for utility i, and c_1 and c_2 are the lower and upper cutoffs, respectively. The proposed classifications were compared to ERSAR classes.

4.2 Results for Real Losses

The results from the quantile-based schemes are presented in 8.3.

Across all quantile-based schemes tested for Real Losses, the tercile-based classification achieved the highest level of agreement with ERSAR performance classes, with an agreement rate close to 0.80. This was followed by the p40–p80 and quartile schemes, while the p20–p80 scheme consistently exhibited the lowest agreement. At first glance, this result might suggest that terciles provide the most accurate approximation of ERSAR's classification. However, a more detailed examination of the contingency tables reveals important qualitative differences between the schemes.

The tercile-based approach produces a very sharp separation between classes, concentrating the majority of ERSAR-Medium utilities within the proposed Medium class. While this contributes to higher agreement, it also introduces abrupt transitions between performance classes that are highly sensitive to shifts in the empirical distribution. Small changes in sector performance can therefore lead to disproportionately large changes in classification outcomes. In contrast, the quartile and p40–p80 schemes generate more gradual transitions between performance classes. Although this comes at the cost of slightly lower agreement with ERSAR, these schemes provide a smoother and more stable classification structure that better reflects continuous variation in RL performance.

An analysis of the proposed class balances further highlights these differences. Both the p20–p80 and p40–p80 schemes tend to produce artificially small Unsatisfactory classes, effectively under-representing poor performance. Conversely, the tercile and quartile schemes yield class distributions that are more consistent with the observed dispersion of RL values, capturing both moderate and extreme performance more effectively. Taken together, these results indicate that while terciles maximize agreement, quartiles offer a more robust and interpretable compromise between agreement and stability.

4.3 Results for Non-Revenue Water

The results from the quantile-based schemes are presented in 8.4.

For Non-Revenue Water, all quantile-based schemes exhibited substantially lower agreement with ERSAR classifications than those observed for Real Losses. This outcome reflects both the fundamental structural difference between ERSAR's NRW indicator, which is defined as a percentage of system input, and the proposed NRW intensity measures, which are expressed in physical units, as well as the highly skewed regulatory assessment of NRW performance. In the reference year, ERSAR classified approximately 63% of utilities as Unsatisfactory, indicating that poor NRW performance is widespread across the sector. Under these conditions, perfect alignment between relative, distribution-based classifications and ERSAR's absolute performance assessment should not be expected.

Among the tested schemes, terciles again achieved the highest agreement with ERSAR, although the agreement remained below 0.50. Quartiles and the p40–p80 scheme produced more balanced class distributions, whereas the p20–p80 scheme excessively compressed the Unsatisfactory class, reassigning a large fraction of utilities to the Medium category. Across all schemes, purely distribution-based thresholds diverged systematically from ERSAR's class balance by allocating a substantially higher proportion of utilities to Medium performance, despite the regulator identifying nearly two-thirds of the sector as failing to meet minimum NRW standards.

This systematic divergence indicates that, when assessed using the proposed NRW indicators, a large number of utilities cluster just above ERSAR's minimum acceptable performance threshold. As a result, a purely data-driven classification risks under-penalizing poor performance by normalizing widespread inefficiency within the sector and mechanically redistributing

utilities into acceptable categories. In a context where Unsatisfactory performance is the dominant regulatory outcome, relative classification schemes inherently weaken the regulatory signal. These findings therefore confirm that, for NRW in particular, a distribution-based approach alone is insufficient to preserve the substantive regulatory meaning of Unsatisfactory performance.

4.4 Hybrid classification scheme

To overcome the limitations identified in the purely quantile-based schemes, a hybrid classification approach was adopted. This approach explicitly combines regulatory grounding with data-driven discrimination. The Unsatisfactory threshold is anchored to ERSAR's own Unsatisfactory group, thereby preserving the regulatory interpretation of minimum acceptable performance, while the distinction between Good and Medium performance is determined empirically within the compliant group.

For each indicator and unit of measurement, the Medium/Unsatisfactory cutoff was defined using a lower-tail quantile of the ERSAR-Unsatisfactory group. A 25th percentile threshold was selected for NRW, reflecting the greater dispersion and weaker alignment between ERSAR and the proposed NRW indicators, while a more conservative 10th percentile threshold was adopted for RL, where alignment with ERSAR is stronger. The Good/Medium cutoff was defined as the median of the ERSAR-compliant group in both cases.

This construction guarantees that no utility classified as Unsatisfactory by ERSAR can be labelled Good under the proposed scheme, while still allowing a meaningful and data-driven distinction between Good and Medium performance. By embedding regulatory constraints within a flexible empirical framework, the hybrid approach reconciles interpretability, stability, and policy relevance in a way that neither absolute benchmarks nor purely relative thresholds can achieve on their own.

During the estimation of hybrid thresholds, it was observed that for some indicator–unit combinations the number of available observations was insufficient to reliably compute the required quantiles, as it would yield statistically unstable results. In such cases, a quartile-based fallback was applied. Quartiles were selected as the fallback scheme because they provide the most statistically stable and assumption-light thresholds under limited sample sizes, representing a conservative and robust approximation. This fallback was used only where necessary and explicitly identified in the cutoff tables.

4.5 Results from hybrid scheme

The results from the hybrid scheme for Real Losses and Non-Revenue Water are presented in 8.5 and 8.6, respectively.

The application of the hybrid classification scheme resulted in clearly differentiated performance distributions for the two indicators under analysis. For Non-Revenue Water (NRW), 26.2% of utilities were classified as exhibiting Good performance, 19.9% as Medium, and 53.9% as Unsatisfactory. In contrast, the Real Losses (RL) indicator displayed a more balanced distribution, with 34.5% of utilities classified as Good, 34.5% as Medium, and 31.1% as Unsatisfactory.

These outcomes closely reflect the underlying sectoral realities and regulatory assessments. In the case of NRW, the predominance of Unsatisfactory classifications is consistent with ERSAR's findings and highlights the structural challenges faced by a large share of utilities in controlling commercial and physical losses. The hybrid scheme therefore does not artificially improve sectoral performance through relative thresholds, but instead preserves the signal that NRW remains a critical area of concern. For RL, the more even distribution across performance classes captures the presence of gradual improvements and heterogeneity among utilities, aligning with the stronger correspondence observed between the proposed RL indicator and ERSAR's classifications.

The contingency analyses further confirm the effectiveness of the hybrid approach. The scheme preserves ERSAR's Unsatisfactory classification for the large majority of utilities, ensuring that entities failing to meet minimum regulatory standards are not reclassified as acceptable performers. At the same time, it reduces misclassification among borderline utilities by introducing a data-driven separation between Good and Medium performance within the ERSAR-compliant group. Importantly, the hybrid approach avoids the class collapse phenomena observed in some purely quantile-based schemes, particularly those relying on extreme-tail cutoffs, which tended to concentrate a disproportionate number of utilities into a single performance class.

Overall, the hybrid classification achieves a balance between regulatory fidelity and empirical discrimination, producing performance distributions that are both interpretable and consistent with observed sector dynamics.

5 Recommended Indicators and Performance Thresholds

This section outlines the normative recommendations for the regulatory framework, synthesizing the findings from the exploratory data analysis and the statistical modelling performed in the previous steps.

5.1 Selection of Indicators

To address the identified limitations of percentage-based indicators and rigid density classifications, it is recommended that the regulator transitions to the following intensity-based indicators:

- **For Real Losses (RL):**
  - **Retail Systems:** Adoption of the **Combined Real Loss Index (CRLI)** as the primary metric. The analysis confirms that CRLI effectively mitigates the systematic bias associated with network density and area typology by geometrically smoothing the interaction between losses per connection and losses per mains length.
  - **Bulk Systems:** Maintenance of **Real Losses per mains length ($m^3/km \cdot day$)**. In these systems, losses are strictly driven by infrastructure extent rather than service connections, and the distribution remains strongly right-skewed.
- **For Non-Revenue Water (NRW):**
  - Replacement of the percentage-based indicator with **Normalized NRW**.
  - **High-Density Systems (≥ 20 connections/km):** Use **NRW per service connection ($L/connection \cdot day$)**.
  - **Low-Density (< 20 connections/km) and Bulk Systems:** Use **NRW per mains length ($m^3/km \cdot day$)**.
  - This conditional approach ensures that the assessment reflects the dominant structural driver of losses (connection density vs. network extent) for each utility profile.

5.2 Performance Thresholds

The definition of performance classes is critical for the policy relevance of benchmarking exercises. The analysis demonstrated that purely data-driven classification schemes, such as those based on terciles or quartiles, while statistically robust, risk normalising sector-wide inefficiency by mechanically assigning a fixed proportion of utilities to "Good" performance regardless of absolute performance levels. This limitation is particularly problematic in sectors where overall performance is low, as relative thresholds may weaken the regulatory meaning of minimum acceptable service.

Therefore, the following approach is recommended.

- **Adoption of a Hybrid Classification Scheme**: Performance thresholds should not be defined exclusively on a relative basis. Instead, the Unsatisfactory threshold should be explicitly anchored to ERSAR's regulatory classification, thereby preserving the interpretation of minimum acceptable performance. In practice, this is achieved by defining the Medium/Unsatisfactory cutoff using a lower-tail quantile of the group classified as Unsatisfactory by ERSAR (e.g., the 25th percentile for NRW and the 10th percentile for Real Losses). Conversely, the distinction between Good and Medium performance should be derived empirically within the ERSAR-compliant group, using a stable central statistic such as the median. This hybrid construction combines regulatory firmness with data-driven discrimination among compliant utilities, avoiding both arbitrary absolute benchmarks and purely relative normalisation.
- **Calibration and Fixation of Reference Thresholds**: To ensure longitudinal comparability, hybrid performance thresholds should be calibrated on a fixed reference period and not recalculated on a year-by-year basis. Re-estimating thresholds annually would reintroduce sensitivity to distributional shifts and undermine the regulatory interpretation of performance classes, particularly in contexts where sector-wide performance evolves over time. Fixing thresholds after calibration allows observed changes in classification to be interpreted as genuine performance improvements or deteriorations rather than statistical artefacts, thereby strengthening the framework's suitability for regulatory monitoring and policy evaluation.
- **Treatment of Units with Limited Observations**: In cases where insufficient data prevent the reliable estimation of hybrid thresholds for specific units of measurement, a transparent fallback strategy should be applied. Quartile-based thresholds provide the most statistically stable and assumption-light alternative under limited sample sizes and are therefore recommended as a conservative fallback. The use of quartile-based thresholds in these cases should be

interpreted strictly as a technical safeguard rather than an alternative classification methodology. This fallback is applied only when hybrid thresholds cannot be estimated reliably and serves to preserve methodological consistency while avoiding unstable or undefined cutoffs under limited sample sizes.

Overall, the hybrid thresholding framework offers a balanced and policy-relevant solution. By combining regulatory anchoring with empirical stability, it preserves the substantive meaning of performance categories while enabling meaningful differentiation among utilities. Although more complex to implement than purely relative schemes, the resulting gains in interpretability, robustness, and regulatory coherence justify its adoption for comparative analysis and longitudinal benchmarking.

6 Discussion

This section critically analyzes the classification results obtained using the proposed hybrid framework compared to the official ERSAR classification and discusses the limitations of the recommended solution.

6.1 Comparison with ERSAR Classification

The application of the Hybrid Classification Scheme highlights distinct patterns for the two indicators, correcting specific distortions present in the current regulatory methodology:

- **Non-Revenue Water:** The proposed framework reveals a more critical scenario than the current percentage-based assessment. Under the hybrid scheme, approximately **53.9%** of utilities are classified as **"Unsatisfactory"** , a dominant share that aligns with ERSAR's identification of problematic cases but refuses to artificially grade them up. Unlike pure quantile schemes, which might misclassify poor performers as "Medium" simply because the sector average is low, the hybrid approach maintains a rigorous standard. However, it provides better differentiation at the top, identifying a clear "Good" performance group (26.2%) that is truly efficient in terms of volume per connection/km, rather than just having high system input volumes that dilute percentages.

- **Real Losses:** The agreement with ERSAR is stronger for Real Losses, but the proposed classification offers superior granularity. The resulting distribution is remarkably balanced (**≈34.5% Good, ≈34.5% Medium, ≈31.1% Unsatisfactory**). This contrasts with the often binary nature of regulatory assessments. The use of CRLI allows for a fairer assessment of utilities in the "middle ground"—often classified as "Good" by ERSAR due to low pressure or specific density traits—by reassigning them to "Medium" where operational improvements are still necessary.

6.2 Limitations and Future Improvements

While the proposed framework offers improvements in fairness and comparability, it entails specific limitations:

1. **Structural Discontinuity:** The NRW indicator relies on a hard cut-off at 20 connections/km to switch units (from m^3/km to L/connection). This creates a discontinuity where utilities with marginally different densities (e.g., 19.9 vs. 20.1) are assessed on different scales. Future iterations should explore a "Combined NRW Index" (similar to CRLI) to smooth this transition.

2. **Aggregation Masking:** NRW is an aggregate of Real Losses, Apparent Losses, and Authorized Unbilled Consumption. Classifying a utility as "Unsatisfactory" in NRW does not immediately reveal the root cause. A utility driven by high leakage requires different interventions than one driven by commercial theft or metering errors. Future work should integrate component-specific ratios (UA_{new} and AL_{new}) as a secondary diagnostic layer to support targeted recommendations.

3. **Dependency on Reference Classification:** The Hybrid scheme's robustness relies on the assumption that ERSAR's "Unsatisfactory" group is a valid proxy for minimum acceptability. If the regulator's baseline criteria were to be fundamentally flawed, the anchored thresholds would propagate that bias. However, this is a necessary trade-off to ensure continuity with existing regulatory standards and avoid the pitfalls of purely relative benchmarking mentioned in the analysis.

7 Conclusion

This study examined alternative indicators and classification frameworks for Non-Revenue Water and Real Losses using regulatory data from the Portuguese water sector. By adopting intensity-based measures, the analysis improves comparability across utilities with heterogeneous structural and operational characteristics.

The empirical results confirm that Real Losses are primarily driven by physical and infrastructural factors, while NRW reflects a broader combination of physical losses, commercial inefficiencies, and management-related aspects. These differences justify the use of distinct indicators and classification strategies for the two dimensions.

The evaluation of alternative classification schemes shows that purely quantile-based approaches, while statistically robust, may weaken the regulatory meaning of minimum acceptable performance by normalising sector-wide inefficiency. This limitation is particularly evident for NRW. To address this issue, a hybrid classification framework was proposed, anchoring the Unsatisfactory category to ERSAR's regulatory classification while empirically distinguishing Good and Medium performance within the compliant group.

By fixing the resulting thresholds over time, the hybrid approach ensures longitudinal comparability and allows observed changes in classification to be interpreted as genuine performance evolution rather than artefacts of shifting empirical distributions. Although more complex to implement than purely relative schemes, the hybrid framework offers superior interpretability, stability, and policy relevance, making it well suited for regulatory benchmarking and performance monitoring.

References

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8 Appendix

8.1 Additional models for NRW

```
Call:
lm(formula = sqrt(NRW_new) ~ log_len + log_in + I_Ocorrência_de_falhas_no_abastecimento +
    V_Tipologia_da_área_de_intervenção + I_Modelo_de_gestão +
    Tipo_Sistema * log_in + I_Ocorrência_de_falhas_no_abastecimento *
    log_len, data = model_df)

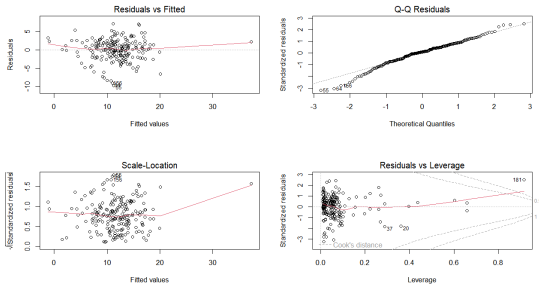
Residuals:
    Min      1Q  Median      3Q     Max
-9.6913 -1.6792  0.2481  1.8473  7.1292

Coefficients:

                                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                                     -49.2988    14.0505  -3.509 0.000562 ***
log_len                                                          -3.2939     0.4350  -7.573 1.52e-12 ***
log_in                                                            4.1068     0.8788   4.673 5.59e-06 ***
I_Ocorrência_de_falhas_no_abastecimento                           1.0089     0.2229   4.527 1.05e-05 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente rural    2.8586     0.6677   4.281 2.94e-05 ***
V_Tipologia_da_área_de_intervençãoArea predominantemente urbana  -1.6644     0.8484  -1.962 0.051250 .
I_Modelo_de_gestãoConcessão (concessão municipal)                 5.9660     1.8963   3.146 0.001920 **
I_Modelo_de_gestãoDelegação (empresa estatal)                     2.0388     2.6354   0.774 0.440119
I_Modelo_de_gestãoDelegação (empresa municipal ou intermunicipal) 6.4490     1.8521   3.482 0.000617 ***
I_Modelo_de_gestãoDelegação (parceria Estado/ municípios)         2.4735     2.2487   1.100 0.272746
I_Modelo_de_gestãoGestão direta (serviço municipal)              10.3476     1.8522   5.587 7.89e-08 ***
I_Modelo_de_gestãoGestão direta (serviço municipalizado ou intermunicipalizado)  8.4124  1.9348  4.348 2.23e-05 ***
Tipo_SistemaBaixa                                                -6.2346    12.8151  -0.487 0.627169
log_in:Tipo_SistemaBaixa                                          1.1259     0.7578   1.486 0.139023
log_len:I_Ocorrência_de_falhas_no_abastecimento                  -0.1412     0.0458  -3.082 0.002361 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.038 on 191 degrees of freedom
Multiple R-squared:  0.659,     Adjusted R-squared:  0.634
F-statistic: 26.36 on 14 and 191 DF,  p-value: < 2.2e-16
```

(a) `m2` summary



(b) `m2` plots

Figure 6: Model `m2`

```
Call:
lm(formula = log(NRW_new) ~ log_len + log_in + I_Ocorrência_de_falhas_no_abastecimento +
    V_Tipologia_da_área_de_intervenção + Tipo_Sistema + I_Modelo_de_gestão +
    Tipo_Sistema * log_in + I_Ocorrência_de_falhas_no_abastecimento *
    log_len, data = model_df)

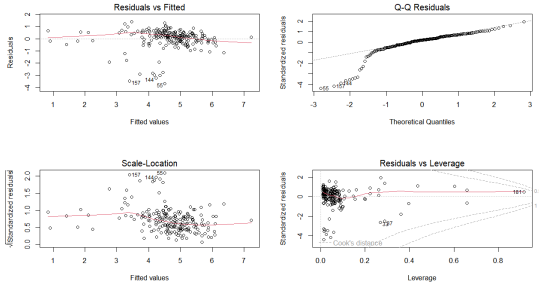
Residuals:
    Min      1Q  Median      3Q     Max
-3.6780 -0.2046  0.1579  0.4865  1.3951

Coefficients:

                                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                                    -9.439401   3.897208  -2.422 0.016365 *
log_len                                                        -0.801173   0.120642  -6.641 3.15e-10 ***
log_in                                                          0.920450   0.243765   3.776 0.000213 ***
I_Ocorrência_de_falhas_no_abastecimento                         0.031752   0.061814   0.514 0.608078
V_Tipologia_da_área_de_intervençãoArea predominantemente rural  0.610454   0.185206   3.296 0.001169 **
V_Tipologia_da_área_de_intervençãoArea predominantemente urbana -0.465127  0.235328  -1.977 0.049537 *
Tipo_SistemaBaixa                                              -0.267124   3.554552  -0.075 0.940174
I_Modelo_de_gestãoConcessão (concessão municipal)               1.835780   0.525985   3.490 0.000599 ***
I_Modelo_de_gestãoDelegação (empresa estatal)                   1.203688   0.730986   1.647 0.101272
I_Modelo_de_gestãoDelegação (empresa municipal ou intermunicipal) 2.006677  0.513720  3.906 0.000130 ***
I_Modelo_de_gestãoDelegação (parceria Estado/ municípios)       0.503088   0.623735   0.807 0.420916
I_Modelo_de_gestãoGestão direta (serviço municipal)             2.527906   0.513759   4.920 1.86e-06 ***
I_Modelo_de_gestãoGestão direta (serviço municipalizado ou intermunicipalizado)  2.248054  0.536661  4.189 4.28e-05 ***
log_in:Tipo_SistemaBaixa                                        0.214749   0.210204   1.022 0.308253
log_len:I_Ocorrência_de_falhas_no_abastecimento                -0.001388   0.012705  -0.109 0.913119
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8425 on 191 degrees of freedom
Multiple R-squared:  0.5509,    Adjusted R-squared:  0.518
F-statistic: 16.74 on 14 and 191 DF,  p-value: < 2.2e-16
```

(a) `m21` summary



(b) `m21` plots

Figure 7: Model `m21`

8.2 Additional models for Real Losses

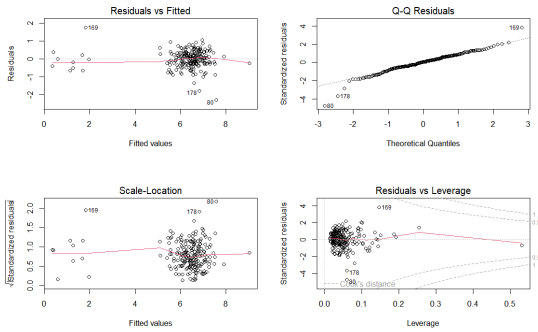
```
Call:
lm(formula = log(RL_new) ~ I_Acessibilidade_física_do_serviço +
    V_Índice_de_conhecimento_infraestrutural + I_Ocorrência_de_falhas_no_abastecimento +
    log_len + log_in + log(V_Rendimentos_tarifários) + I_Adesão_ao_serviço +
    Tipo_Sistema + V_Tipologia_da_área_de_intervenção, data = model_df)

Residuals:
     Min       1Q   Median       3Q      Max
-2.30464 -0.25565  0.04055  0.32157  1.76004

Coefficients:
                                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                                             -5.720365   1.119220  -5.111 7.62e-07 ***
I_Acessibilidade_física_do_serviço                       0.008811   0.005318   1.657 0.099156 .
V_Índice_de_conhecimento_infraestrutural                -0.003697   0.001036  -3.569 0.000451 ***
I_Ocorrência_de_falhas_no_abastecimento                  0.026369   0.009264   2.846 0.004895 **
log_len                                                 -0.631435   0.072903  -8.661 1.76e-15 ***
log_in                                                   1.055583   0.092529  11.408  < 2e-16 ***
log(V_Rendimentos_tarifários)                           -0.413166   0.069186  -5.972 1.09e-08 ***
I_Adesão_ao_serviço                                     -0.008275   0.005022  -1.648 0.100999
Tipo_SistemaBaixa                                        6.715613   0.227773  29.484  < 2e-16 ***
V_Tipologia_da_área_de_intervençãoÁrea predominantemente rural   0.355850   0.106397   3.345 0.000988 ***
V_Tipologia_da_área_de_intervençãoÁrea predominantemente urbana -0.036503   0.139648  -0.261 0.794068
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4987 on 195 degrees of freedom
Multiple R-squared:  0.8697,    Adjusted R-squared:  0.863
F-statistic: 130.2 on 10 and 195 DF,  p-value: < 2.2e-16
```

(a) `m41` summary



(b) `m41` plots

Figure 8: Model `m41`

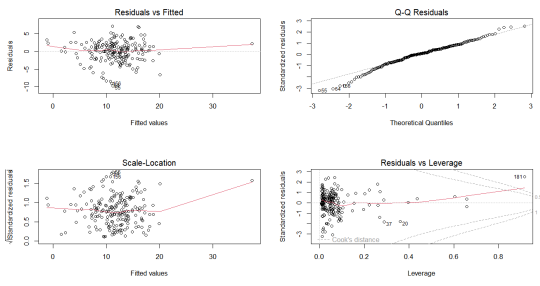
```
Call:
lm(formula = RL_new ~ I_Acessibilidade_física_do_serviço +
    I_Ocorrência_de_falhas_no_abastecimento + log_len + log_in +
    log1p(V_Rendimentos_tarifários) + I_Adesão_ao_serviço +
    I_Densidade_de_ramais + V_Tipologia_da_área_de_intervenção,
    data = subset(model_df, model_df$Tipo_Sistema == "Baixa"))

Residuals:
     Min       1Q   Median       3Q      Max
-1408.08  -221.02   -31.49   174.82  1658.16

Coefficients:
                                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                                             -4173.306    902.933  -4.622 7.09e-06 ***
I_Acessibilidade_física_do_serviço                          7.608      4.864   1.564  0.11949
I_Ocorrência_de_falhas_no_abastecimento                    66.919      8.749   7.649 1.06e-12 ***
log_len                                                  -748.176     93.646  -7.989 1.38e-13 ***
log_in                                                    926.913     92.254  10.047  < 2e-16 ***
log1p(V_Rendimentos_tarifários)                          -261.575     65.302  -4.006 8.93e-05 ***
I_Adesão_ao_serviço                                       -12.965      4.704  -2.756  0.00643 **
I_Densidade_de_ramais                                      -5.989      2.992  -2.002  0.04673 *
V_Tipologia_da_área_de_intervençãoÁrea predominantemente rural   382.565     96.119   3.980 9.87e-05 ***
V_Tipologia_da_área_de_intervençãoÁrea predominantemente urbana  -82.841    131.640  -0.629  0.52992
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 457.7 on 186 degrees of freedom
Multiple R-squared:  0.5907,    Adjusted R-squared:  0.5709
F-statistic: 29.83 on 9 and 186 DF,  p-value: < 2.2e-16
```

(a) `m4b` summary



(b) `m4b` plots

Figure 9: Model `m4b`

8.3 Quantile-based schemes classification results for RL

scheme	p1	p2	agreement
<chr>	<dbl>	<dbl>	<dbl>
terciles	0.3333333	0.6666667	0.8009709
p40_p80	0.4000000	0.8000000	0.7912621
quartiles	0.2500000	0.7500000	0.7233010
p20_p80	0.2000000	0.8000000	0.6601942

(a) Proportion of utilities classified in the same performance category by ERSAR and proposed schemes.

scheme	ERSAR	Good	Medium	Unsatisfactory
<chr>	<chr>	<dbl>	<dbl>	<dbl>
p20_p80	Good	0.46666667	0.53333333	0.00000000
p20_p80	Medium	0.00000000	0.98181818	0.01818182
p20_p80	Unsatisfactory	0.00000000	0.34426230	0.65573770
p40_p80	Good	0.84444444	0.15555556	0.00000000
p40_p80	Medium	0.12727273	0.85454545	0.01818182
p40_p80	Unsatisfactory	0.00000000	0.34426230	0.65573770
quartiles	Good	0.57777778	0.42222222	0.00000000
quartiles	Medium	0.00000000	0.90909091	0.09090909
quartiles	Unsatisfactory	0.00000000	0.22950820	0.77049180
terciles	Good	0.75555556	0.24444444	0.00000000
terciles	Medium	0.03636364	0.74545455	0.21818182
terciles	Unsatisfactory	0.00000000	0.08196721	0.91803279

(b) Conditional distribution of proposed performance classes given ERSAR classes.

scheme	p1	p2	unit	n	cut1	cut2
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>	<dbl>
quartiles	0.2500000	0.7500000	CRLI	196	420.443421	1067.230351
quartiles	0.2500000	0.7500000	m3/km/day	10	1.910470	5.733545
terciles	0.3333333	0.6666667	CRLI	196	535.428730	897.991054
terciles	0.3333333	0.6666667	m3/km/day	10	2.141115	2.904840
p20_p80	0.2000000	0.8000000	CRLI	196	383.167405	1232.511333
p20_p80	0.2000000	0.8000000	m3/km/day	10	1.796450	6.748565
p40_p80	0.4000000	0.8000000	CRLI	196	597.152976	1232.511333
p40_p80	0.4000000	0.8000000	m3/km/day	10	2.173778	6.748565

(c) Empirical threshold values (cutoffs).

scheme	Good	Medium	Unsatisfactory
<chr>	<dbl>	<dbl>	<dbl>
ERSAR	0.4368932	0.2669903	0.2961165
p20_p80	0.2038835	0.5970874	0.1990291
p40_p80	0.4029126	0.3980583	0.1990291
quartiles	0.2524272	0.4951456	0.2524272
terciles	0.3398058	0.3300971	0.3300971

(d) Distribution of utilities across performance classes.

Figure 10: Sensitivity of NRW classifications to alternative empirical threshold schemes.

8.4 Quantile-based schemes classification results for NRW

scheme	p1	p2	agreement
<chr>	<dbl>	<dbl>	<dbl>
terciles	0.3333333	0.6666667	0.4611650
quartiles	0.2500000	0.7500000	0.3932039
p20_p80	0.2000000	0.8000000	0.3883495
p40_p80	0.4000000	0.8000000	0.3786408

(a) Proportion of utilities classified in the same performance category by ERSAR and proposed schemes.

scheme	ERSAR	Good	Medium	Unsatisfactory
<chr>	<chr>	<dbl>	<dbl>	<dbl>
p20_p80	Good	0.56097561	0.3658537	0.07317073
p20_p80	Medium	0.28571429	0.6285714	0.08571429
p20_p80	Unsatisfactory	0.06923077	0.6615385	0.26923077
p40_p80	Good	0.73170732	0.1951220	0.07317073
p40_p80	Medium	0.54285714	0.3714286	0.08571429
p40_p80	Unsatisfactory	0.26153846	0.4692308	0.26923077
quartiles	Good	0.58536585	0.3414634	0.07317073
quartiles	Medium	0.45714286	0.4000000	0.14285714
quartiles	Unsatisfactory	0.10000000	0.5692308	0.33076923
terciles	Good	0.70731707	0.1707317	0.12195122
terciles	Medium	0.54285714	0.2571429	0.20000000
terciles	Unsatisfactory	0.16153846	0.4000000	0.43846154

(b) Conditional distribution of proposed performance classes given ERSAR classes.

scheme	p1	p2	unit	n	cut1	cut2
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>	<dbl>
quartiles	0.2500000	0.7500000	L/conn/day	185	100.909585	222.581023
quartiles	0.2500000	0.7500000	m3/km/day	21	2.599505	8.000804
terciles	0.3333333	0.6666667	L/conn/day	185	117.342748	189.483624
terciles	0.3333333	0.6666667	m3/km/day	21	3.147989	6.117738
p20_p80	0.2000000	0.8000000	L/conn/day	185	92.654527	255.883045
p20_p80	0.2000000	0.8000000	m3/km/day	21	2.352884	11.232688
p40_p80	0.4000000	0.8000000	L/conn/day	185	126.114429	255.883045
p40_p80	0.4000000	0.8000000	m3/km/day	21	3.612675	11.232688

(c) Empirical threshold values (cutoffs).

scheme	Good	Medium	Unsatisfactory
<chr>	<dbl>	<dbl>	<dbl>
ERSAR	0.1990291	0.1699029	0.6310680
p20_p80	0.2038835	0.5970874	0.1990291
p40_p80	0.4029126	0.3980583	0.1990291
quartiles	0.2572816	0.4951456	0.2475728
terciles	0.3349515	0.3300971	0.3349515

(d) Distribution of utilities across performance classes.

Figure 11: Sensitivity of NRW classifications to alternative empirical threshold schemes.

8.5 Hybrid scheme classification results for RL

Class <fctr>	n <int>	pct <dbl>
Good	71	0.3446602
Medium	71	0.3446602
Unsatisfactory	64	0.3106796

(a) Proportion of utilities classified in each performance category under the hybrid schemes.

ERSAR <chr>	Proposed <fctr>	n <int>	pct <dbl>
Good	Good	69	0.76666667
Good	Medium	21	0.23333333
Medium	Good	2	0.03636364
Medium	Medium	44	0.80000000
Medium	Unsatisfactory	9	0.16363636
Unsatisfactory	Medium	6	0.09836066
Unsatisfactory	Unsatisfactory	55	0.90163934

(b) Conditional distribution of hybrid performance classes given ERSAR classifications.

RL_new_unit <chr>	n_non_unsat <int>	cut1 <dbl>	n_unsat <int>	cut2 <dbl>	n_all <int>	q25 <dbl>	q75 <dbl>	final_cut1 <dbl>	final_cut2 <dbl>
CRLI	136	541.9227	60	953.1186	196	420.44342	1067.230351	541.92274	953.118633
m3/km/day	9	*NA*	1	*NA*	10	1.91047	5.733545	1.91047	5.733545

(c) Threshold values (cutoffs).

Figure 12: RL classification results under the hybrid threshold scheme.

8.6 Hybrid scheme classification results for NRW

Class <fctr>	n <int>	pct <dbl>
Good	54	0.2621359
Medium	41	0.1990291
Unsatisfactory	111	0.5388350

(a) Proportion of utilities classified in each performance category under the hybrid schemes.

ERSAR <chr>	Proposed <fctr>	n <int>	pct <dbl>
Good	Good	24	0.5853659
Good	Medium	11	0.2682927
Good	Unsatisfactory	6	0.1463415
Medium	Good	14	0.4000000
Medium	Medium	8	0.2285714
Medium	Unsatisfactory	13	0.3714286
Unsatisfactory	Good	16	0.1230769
Unsatisfactory	Medium	22	0.1692308
Unsatisfactory	Unsatisfactory	92	0.7076923

(b) Conditional distribution of hybrid performance classes given ERSAR classifications.

NRW_new_unit <chr>	n_non_unsat <int>	cut1 <dbl>	n_unsat <int>	cut2 <dbl>	n_all <int>	q25 <dbl>	q75 <dbl>	final_cut1 <dbl>	final_cut2 <dbl>
L/conn/day	64	99.419769	121	129.633	185	100.909585	222.581023	99.419769	129.632964
m3/km/day	12	4.662116	9	*NA*	21	2.599505	8.000804	4.662116	8.000804

(c) Threshold values (cutoffs).

Figure 13: NRW classification results under the hybrid threshold scheme.